

What RAG is, and how it works

DATA 443/643 · Week 14, note 14.1

Week 14

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2026 L10 and Lewis et al. 2020. Original slides; those sources are not copied.

Lecture plan (first time seeing RAG)

- 1 What it is, and why weights are not enough
- 2 Architecture: index, retrieve, generate
- 3 Formulas: cosine, top-k, next-token, Lewis's sum
- 4 Pros, cons, and four failure modes
- 5 Worked ranking: cosine 1 / 0.707 / 0

Key goal: define RAG, draw the boxes, write the formulas, and name one failure retrieval cannot fix.

What RAG is

- A frozen LLM answers from weights. Those weights are a snapshot.
- RAG: look up passages from a collection you control, then generate from them.
- Retrieval finds text. Augmented means it goes in the prompt. Generation is still next-token.

RAG: look up passages, then write the answer from them



Why bother

Closed book (weights only)

Fast. Stale. No chunk id to cite. Fluent guesses when the fact is missing.

RAG (index + generator)

Edit a PDF and re-index. Cite [d1]. Keep a modest LLM plus a large library.

Three ways to give a model new facts

Three ways to give a model new facts. RAG edits an index.

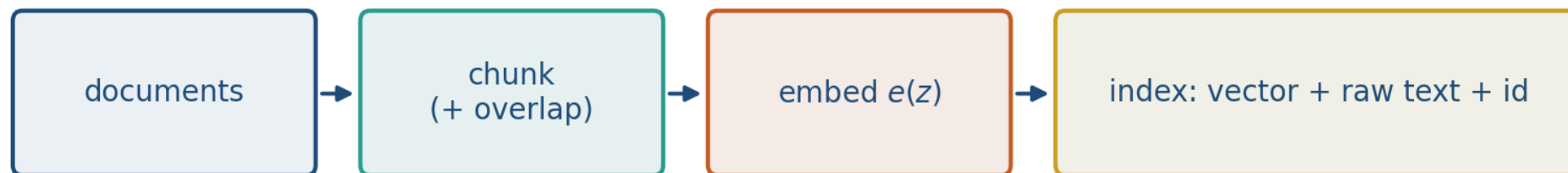
method	updates by	cites?	main cost
fine-tune / LoRA	changing weights	no	retrain
long context	pasting everything	weak	tokens + middle
RAG	editing the index	yes, if you store ids	retrieve + k

Fine-tune changes weights. Long context pastes everything. RAG retrieves a few chunks.

Architecture: the transformer does not change

RAG architecture: the transformer is unchanged; the prompt is not

Offline (build the index)



Online (every question)



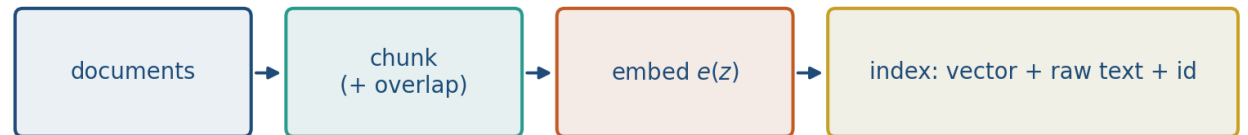
Offline: chunk, embed, store vector+text+id. Online: embed the query, top-k, stuff, generate.

How a question runs

- Embed the query with the same map as the chunks.
- Keep the k nearest passages. Stuff them with ids into a prompt.
- Instruct: answer from the passages, cite, or say the context is missing.

RAG architecture: the transformer is unchanged; the prompt is not

Offline (build the index)



Online (every question)



Retrieve by cosine

$$\cos(e(q), e(z)) = \frac{e(q)^\top e(z)}{\|e(q)\| \|e(z)\|}$$

$$\hat{z}_{1:k} = \text{TopK}_z \cos(e(q), e(z))$$

Lab 14: e is bag-of-words counts. A product uses a neural embedding.

Same embedder for questions and chunks. Mixing maps breaks ranking.

Generate from the stuffed prompt

$$p_{\theta}(y \mid q, \hat{z}_{1:k}) = \prod_t p_{\theta}(y_t \mid y_{<t}, x)$$

$x = \text{Prompt}(q, \hat{z}_{1:k})$
 . RAG changed $\mathcal{X}$, not the net.

Lewis et al.:

$$p(y \mid q) = \sum_z p_{\eta}(z \mid q) p_{\theta}(y \mid q, z)$$

The lab does hard top-k stuffing, not the full sum.

What it is good and bad at

Positive

Update facts by editing the index. Cite ids. Frozen generator. Private PDFs stay out of the weights.

Negative

A miss cannot be fixed by a prettier prompt. Latency. Chunking and k are method. Lost in the middle. Unfaithful citations.

Four failures to name

Miss

Gold never entered top-k. Measure recall@k.

Near-miss

Related junk ranks high; the model over-trusts it.

Ignore

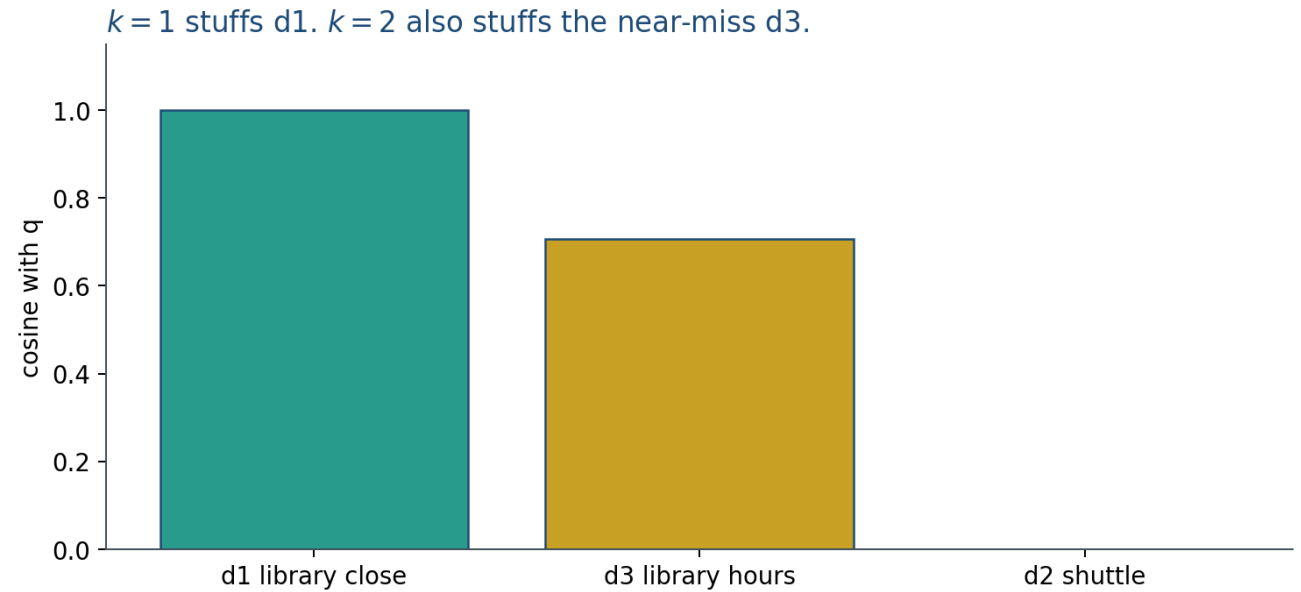
Gold is in the prompt; weights still win. RAFT's job.

Bad cite

Used [d1], wrote [d2]. Unfaithful, like CoT.

k is a method knob

- $k=1$ stuffs d1 (the close-time chunk).
- $k=2$ also stuffs d3 (hours). Recall can rise; the near-miss can mislead.
- Lost in the middle: gold in a long context is often ignored.



Worked cosine on three toy docs

$$\cos(q, d1) = 1, \quad \cos(q, d3) = 1 / \sqrt{2}, \quad \cos(q, d2) = 0$$

$$\begin{aligned} q &= d1 = (1, 1, 0). \\ d3 &= (1, 0, 0). \\ d2 &= (0, 0, 1). \end{aligned}$$

Rank: d1, d3, d2. Lab 14 prints this ranking. No GPU.

Recall@k needs gold ids. Attribution needs the cited id to contain the claim.

Two weeks, two jobs

10.3 RAFT

A training recipe: learn to use retrieved docs, distractors included. Weights change.

14.1 this hour

The inference stack, taught from scratch: what, why, boxes, formulas, then the ranking.

What to take from this lecture

Definition

Retrieve passages, stuff them, generate. Not a new net.

Two metrics

Retrieval recall@k, then answer + attribution.

Cite ids

Vectors without raw text cannot be cited.

Lab 14

Five AU-toy snippets. Cosine, then a citation.